

ST HELENA, Saint Helena

WMO: 619010

Lat: 15.9422S

Lon: 5.6676W

Elev: 1430

StdP: 13.95

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	56.5	57.1	48.7	53.5	61.4	49.9	56.1	61.4	31.5	60.1	29.2	59.7	14.4	130	0.600

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	6.7	74.9	67.0	73.8	66.7	72.9	66.4	69.7	71.4	69.1	71.0	68.4	70.5	13.2	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.2	113.0	70.3	68.4	110.0	69.8	67.8	107.6	69.5	34.6	71.6	34.0	71.1	33.6	70.6	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.3	24.2	22.6	54.2	78.5	1.5	2.9	53.1	80.6	52.3	82.3	51.4	84.0	50.4	86.1		
(5)			51.0	71.7	1.4	2.1	50.0	73.3	49.2	74.5	48.4	75.7	47.4	77.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	65.1	67.8	69.8	70.3	69.0	67.0	64.0	62.1	60.8	60.8	61.4	62.9	65.1	(6)
(7)		DBStd	3.72	1.50	1.38	1.34	1.41	1.65	1.49	1.53	1.52	1.33	1.43	1.51	1.70	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	582	1	0	0	0	3	36	90	129	126	111	66	20	(9)
(10)		CDD50	5500	552	555	629	571	527	420	376	336	324	355	386	469	(10)
(11)		CDD65	606	88	135	164	121	65	7	0	0	0	0	2	24	(11)
(12)		CDH74	82	6	30	36	8	2	0	0	0	0	0	0	0	(12)
(13)		CDH80	1	0	0	0	0	1	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	14.5	13.5	13.0	13.7	13.8	12.8	13.8	14.7	16.3	16.4	16.5	15.5	14.3	(14)
(15)	Precipitation	PrecAvg	19.6	1.4	1.6	2.0	1.7	1.6	2.4	2.2	2.2	1.5	1.0	0.7	1.1	(15)
(16)		PrecMax	28.9	3.3	4.7	4.1	3.2	4.5	5.2	4.9	4.0	3.2	2.4	1.6	4.3	(16)
(17)		PrecMin	13.3	0.3	0.4	0.7	0.4	0.4	0.1	0.2	0.6	0.2	0.2	0.1	0.3	(17)
(18)		PrecStd	3.8	0.8	0.8	0.9	0.9	0.9	1.1	1.1	0.8	0.7	0.6	0.4	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.7	76.5	76.5	75.2	73.2	70.1	68.4	66.5	66.7	68.2	69.7	71.8	(19)
(20)			MCWB	65.5	66.9	68.0	66.7	64.9	62.4	60.3	59.4	59.8	60.8	61.6	62.7	(20)
(21)		2%	DB	72.9	74.7	74.9	73.6	71.7	68.7	66.6	65.2	65.1	66.1	67.7	70.2	(21)
(22)			MCWB	64.9	66.8	67.6	66.4	64.5	61.8	59.8	58.9	59.1	60.0	61.0	62.7	(22)
(23)		5%	DB	71.8	73.7	73.8	72.6	70.7	67.5	65.6	64.1	63.9	64.8	66.5	69.2	(23)
(24)			MCWB	64.8	66.7	67.3	66.2	64.2	61.3	59.5	58.5	58.7	59.5	60.5	62.6	(24)
(25)		10%	DB	70.7	72.6	72.8	71.6	69.6	66.5	64.6	63.1	62.9	63.7	65.4	68.1	(25)
(26)			MCWB	64.6	66.6	67.2	66.0	63.9	61.0	59.2	58.2	58.5	59.3	60.2	62.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.2	70.2	71.0	70.0	67.5	65.0	62.6	61.2	61.7	62.6	63.5	66.2	(27)
(28)			MCDWB	70.6	72.0	72.5	71.2	69.9	67.2	64.9	63.4	63.9	65.3	66.2	68.0	(28)
(29)		2%	WB	67.3	69.3	70.0	69.1	66.5	63.7	61.6	60.4	60.6	61.4	62.4	64.9	(29)
(30)			MCDWB	69.7	71.3	71.7	70.6	69.1	66.2	64.0	62.5	62.4	63.9	65.1	67.1	(30)
(31)		5%	WB	66.6	68.7	69.3	68.3	65.8	62.9	61.0	59.9	60.0	60.8	61.8	64.3	(31)
(32)			MCDWB	68.9	70.8	71.2	70.2	68.5	65.5	63.5	62.1	62.0	63.0	64.4	66.6	(32)
(33)		10%	WB	66.1	68.1	68.7	67.6	65.2	62.2	60.4	59.3	59.4	60.1	61.1	63.6	(33)
(34)			MCDWB	68.4	70.3	70.8	69.8	68.0	65.0	63.1	61.6	61.5	62.4	63.7	66.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.2	7.1	6.7	6.3	6.4	6.0	5.7	5.3	5.3	5.7	6.2	6.8	(35)
(36)			MCDBR	8.2	8.1	7.5	7.1	7.3	6.9	6.7	6.0	6.2	6.6	7.3	7.6	(36)
(37)			MCWBR	3.7	3.7	3.7	3.9	4.4	4.5	4.5	4.2	4.1	3.8	4.0	4.1	(37)
(38)		5% WB	MCDWB	7.2	6.9	6.4	5.9	6.6	6.1	5.9	5.4	5.5	5.9	6.5	6.8	(38)
(39)			MCWBR	3.7	3.7	3.5	3.7	4.3	4.5	4.5	4.2	4.1	3.8	4.0	4.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.344	0.342	0.345	0.329	0.316	0.311	0.330	0.355	0.407	0.379	0.335	0.335	(40)	
(41)		taud	2.585	2.616	2.618	2.657	2.665	2.650	2.574	2.483	2.298	2.415	2.586	2.601	(41)	
(42)		Ebn at Noon	317	315	307	301	295	291	287	289	284	300	318	320	(42)	
(43)		Edh at Noon	34	32	31	28	26	26	29	33	43	39	33	33	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2198	2217	2047	1750	1512	1312	1332	1396	1495	1597	1819	2033	(44)	
(45)		RadStd	137	75	97	104	67	65	104	101	95	110	130	125	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	+0.36	+0.68	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page